

EDUCATION

- **École Polytechnique** Sept 2024 - June 2027
Bachelor's degree in Mathematics and Computer Science GPA 4.19/4.3
- **Tudor Vianu National High School of Computer Science** Sept 2020 - June 2024
Baccalaureate Diploma 9.73 (Romanian, Mathematics, Physics)

EXPERIENCE

- **Arago, internship - Software Team** June - August 2026
 - Implementing a compute dispatcher on a cluster of ARM and RISC-V embedded cores, fully written in baremetal C and C++.
 - Debugged low level device drivers.
 - Implemented compilation, simulation and testing infrastructure for my project.
- **Arago, internship - Optics Team** June - August 2025
 - Modeling time response of the up-conversion phenomenon. Correlated simulation to experimental results. Showed how the system parameters could be tuned to go from response time of s to an order of μs .
 - Improved prototyping efficiency by creating a Python module for automatic data acquisition through UART and I²C, using the optics lab equipment.
- **Black Shell, internship** June-July 2023
 - One of the first full time employees at the beginning of the startup.
 - Learned Golang and used it to implement key manager APIs for Wireguard and OpenVPN which shipped with the company's main product.

EXTRACURRICULARS

- **IP Paris Formula Student Racing - Electronics Team** 2024-present
 - Working on the Break System Plausibility Device (BSPD), a non-programmable, system critical device that shuts down power when breaking and accelerating at the same time.
 - Designed the "ready to drive" circuit, responsible for powering on the car.
 - Ensured both boards fit within the rule's specifications.
- **CSC - CS Committee of the Polytechnique Bachelor** 2025-2026
 - Vice President; coordinating a 10-member team to organize seminars, competitions, hackathons and other events.
 - Presented an "Introduction to Diffusion" ([link](#)), a seminar aimed at giving some intuition for how diffusion works while also going into the mathematics of DDPM.

PROJECTS

- **Implementation of QR-DQN for CleanRL** 2026
Lab Reserch Project under the supervision of Dr. Simo Alami
 - Implemented the Qunatile Regression DQN algorithm ([Dabney et al. 2017](#)), and contributed it to the CleanRL open source library.
 - Matched the performance reported in the original paper on 5 games of the Atari 2600 benchmark.
 - Wrote a report and held an oral defense explaining my work.
- **Swarm Rescue Competition** 2025
Writing a controller to coordinate a swarm of drones in a simulated environment.
 - Part of a team of 3 students participating in the **Swarm Rescue Competition**
 - Designed a state machine controller for high level decision making
 - Implemented PID controller for the drone's motion.
 - Merged the simulated IMU and GPS measurements to reduce noise in drone position.
 - **source code**
- **DiffEdit** 2025
Implementation of an image editing method using diffusion.

- Made extensive use of the PyTorch and Hugging Face diffusers libraries to write my own diffusion loop as described in the **paper**.
 - **source code**
- **NSS Space Settlement Contest** **Grand Prize**
*Designed a self-sufficient space colony detailed in a **50 page article**.* **2024**
 - Won the Grand Prize out of 1000+ teams.
 - Presented at the 2024 International Space Development Conference.
 - Responsible for the structural design and analysis.
 - Created 3D models and renders in Blender and Fusion360.
 - **Optics Unravelled** **2023**
Built an open source interactive optics simulator & lesson platform.
 - Designed the website to be used as a teaching tool by my high school. (**opticsunravelled.com**)
 - Full stack application in Node.js. (**source code**)
 - **Polytechnic University of Bucharest Cube2Space National Contest** **1st Prize**
Design of a pocket cube to be launched up to 9km on the LUNA rocket. **2022**
 - As leader, managed and organised the ViaSpace team.
 - The pocket cube investigated dampening methods to mitigate the effects of vibrations during launch.
 - Created CAD models in Fusion360, used for rendering and 3D printing of the pocket cube.
 - **ESA Moon Camp Pioneers** **3rd Prize**
Design and 3D modeling of a lunar base in Fusion 360. **2021**
 - In charge of designing the airlock, leisure and other areas of the moon base.
 - Created 3D models and renders of these areas in Fusion360.

HONORS & AWARDS

- **International Zautykov Olympiad (Physics)** - Bronze medal **2023**
- **National Physics Olympiad** - Silver Medal **2022**
- **National Informatics Olympiad** - Bronze Medal **2021 & 2022**
- **International Astronomy and Astrophysics Competition** - Bronze Medal **2022**
- **Unbreakable Hacking Challenge Romania (Team Section)** - 1st Prize **2022**
- **National Astronomy and Astrophysics Olympiad** - Honorable Mention **2021**

SKILLS

Languages: Romanian - native
 English - C2
 French - C1
 German - A1

Computer: C / C++ / C# / Python / Golang / JavaScript / HTML / Fusion360 / Blender / KiCad